

Business Insights

Business Insights & AI Recommendations

Automated analysis based on Gold Layer data | Powered by Databricks Analytics

Revenue Performance

Total Revenue is aggregated from `fact_sales` (Gold Layer). The Olist e-commerce platform shows strong revenue concentration in Southeast Brazil.

Key Findings:

- Revenue follows a clear growth trajectory from 2016 to 2018
- Monthly peaks observed toward Q4 (holiday shopping)
- Average Order Value reflects a healthy mid-range consumer basket

Recommendation: Focus promotional campaigns in Q3 to boost Q4 momentum. Consider volume discounts to increase AOV.

Geographic Opportunities

SP (São Paulo) dominates revenue as the largest Brazilian e-commerce market, followed by RJ and MG.

Key Findings:

- Top 3 states (SP, RJ, MG) represent the majority of all orders
- Northern and Northeastern states show lower penetration but high growth potential
- City-level analysis reveals São Paulo city as the single largest revenue hub

Recommendation: Invest in logistics infrastructure in Northeast (BA, CE, PE) to capture untapped demand. Targeted marketing in secondary cities could yield high ROI.

Customer Retention Strategy

Customer Analysis from `seller_kpi` and `dim_customer` reveals purchasing patterns across the platform.

Key Findings:

- Most customers (>90%) are single-order buyers — low repeat purchase rate
- Top 10 customers by revenue represent VIP accounts worth prioritizing
- Customer geographic spread mirrors population distribution

Recommendation:

- Launch loyalty program to increase repeat purchase rate
- Create VIP customer tier for top spenders with exclusive benefits
- Use post-purchase email sequences to drive second orders within 30 days
- Target high-AOV customers with upsell/cross-sell campaigns

Product & Seller Performance

Product catalog spans multiple categories from `dim_product`. Seller network covers multiple Brazilian states.

Key Findings:

- Product catalog has high category diversity (73+ categories)
- Seller concentration in SP mirrors customer concentration
- No direct revenue-to-product mapping in Gold layer — consider enriching `fact_sales` with `order_items` join

Data Enhancement Opportunities:

- Add `product_id` and `seller_id` to `fact_sales` Gold table for richer analysis
- Create `gold.fact_sales_enriched` joining `silver.order_items_dataset`
- Enable category-level revenue attribution
- Enable individual seller performance tracking and ranking

Growth Opportunities

Revenue Growth shows strong YoY improvement across the analyzed period.

Immediate Actions (0-90 days):

- Identify and re-engage customers with 1 order > 6 months ago
- Promote highest-revenue product categories with targeted ads
- Expand seller network in underrepresented states

Medium Term (90-180 days):

- Build predictive churn model using order recency/frequency data
- Implement dynamic pricing for high-demand categories
- Develop regional fulfillment centers to reduce delivery times

Long Term (6-12 months):

- Cross-sell recommendations engine using purchase history
- Seller performance scoring to ensure quality
- Expand to adjacent Latin American markets

Data Architecture Notes

Current Gold Layer Coverage:

Metric	Availabl e	Source
Total Revenue	Yes	<code>fact_sales</code> + <code>sales_summary</code>
Orders by Time	Yes	<code>daily/monthly/yearly_sales</code>
Customer	Yes	<code>dim_customer</code>